

**Hype versus Fundamentals in IPO Return Prediction:
A Machine Learning Analysis of Feature Importance of U.S. IPOs,
2009-2023**

1. Introduction

Initial public offerings (IPOs) attract significant investor attention, yet IPO stocks tend to underperform the broader market over one to five years after listing. Loughran and Ritter (1995) documented that U.S. firms going public between 1970 and 1990 underperformed matched non-issuers by roughly 44% over five years. This long-run underperformance coexists with strong first-day returns, suggesting a gap between early excitement and actual long-run performance.

This project uses machine learning to ask whether that gap is predictable. We study 3,158 U.S. IPOs from 2009 to 2023 and predict returns at four horizons: the day of listing (Day 0), the day after listing (Day 1), two weeks, and one year. All models use the same 17 pre-IPO features at every horizon, enabling a direct comparison of how well pre-IPO information predicts performance over time. To measure performance relative to the market, we compute buy-and-hold abnormal returns (BHAR), which subtract the S&P 500 return over the same period from the IPO raw return.

Our central hypothesis is the hype versus health hypothesis: early trading behavior (hype) should predict short-run returns, while pre-IPO structural characteristics (fundamentals) should become more useful at longer horizons. We test this by comparing three models on the same observations: a hype-only model using Day-0 and Day-1 returns, a fundamentals-only model using pre-IPO features, and a full model combining both.

2. Data and Methodology

2.1 Data Sources

The base dataset contains 3,158 U.S. IPOs from 2009 to 2023, sourced from LSEG/Refinitiv Workspace. Each record includes offer price, shares offered, filing and issue

dates, Venture Capital (VC) and Private Equity (PE) backing flags, underwriting (UW) fees, employee count, and Return on Assets (ROA). We supplement this with: (1) daily Volatility Index (VIX) data from FRED; (2) daily S&P 500 prices from FRED for benchmark-adjusted returns; and (3) firm-level features scraped from SEC EDGAR S-1 filings, including burn-to-revenue ratio, R&D intensity, number of risk factors, log employee count, revenue Compound Annual Growth Rate (CAGR), and founder CEO status. Figure 1 shows IPO volume by year.

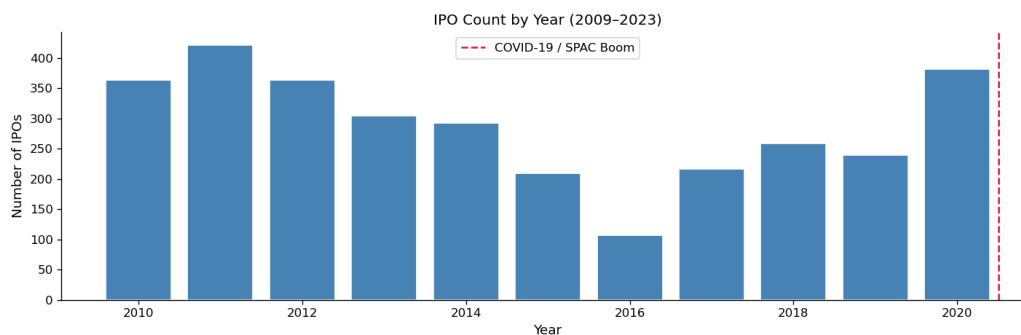


Figure 1: IPO count by year (2009-2023). The dashed red line marks the COVID-19 / SPAC boom starting in 2021.

2.2 Data Cleaning and Feature Engineering

Several data quality issues required careful handling. The raw Day-1 return column was corrupted (Spearman correlation of -0.66 with price-implied returns, 221 entries hard-coded to -1.000); we discarded it and recomputed Day-1 returns from prices, excluding IPOs with offer prices below \$5. The company age field mixed years and days; values above 200 were treated as days and divided by 365. Log proceeds and underwriting fee percentage were median-imputed, as these right-skewed financial variables make the median more robust than the mean (Sterne et al., 2009).

The EDGAR S-1 features are Missing Not At Random (MNAR): our scraper resolved a CIK identifier for 886 of 3,158 companies (28.1%). Following van Buuren (2018) and Sterne et

al. (2009), we created a binary indicator `has_s1_data` and included it as a feature in every model, allowing the algorithm to learn that the two groups are systematically different.

2.3 Outcome Variables

We compute two return measures at each of four horizons (Day 0, Day 1, 2 weeks, and 365 days). Raw returns are the percentage price change from the offer price, winsorized at the 1st and 99th percentile. BHAR subtracts the S&P 500 return over the same period. Table 1 summarizes sample sizes and key return statistics.

Table 1: Return summary by horizon (winsorized at 1st/99th percentile)

Horizon	Day 0	Day 1	2-Week	365-Day
Sample (n)	2,586	1,716	2,266	2,138
Mean Return (%)	-5.92	2.66	23.00	38.65
Median Return (%)	0.00	1.83	4.02	1.76
Median BHAR (%)	-0.04	+2.03	+3.87	-11.96

Note: Returns are in percent. BHAR = raw IPO return minus S&P 500 return over same horizon. Day 0 raw return is offer-to-close on listing day.

2.4 Feature Set and Models

All models use the same 17 pre-IPO features at every horizon, enabling a direct comparison of how predictability changes over time. The features fall into four groups: (1) deal features -- VC-backing, PE-backing, prior LBO flag, log proceeds, UW fee percentage, and filing-to-issue days; (2) firm features -- ROA and company age in years; (3) market and time features -- VIX at the IPO date and IPO year; and (4) S-1 derived features -- burn-to-revenue,

R&D intensity, number of risk factors, log employee count, pre-IPO revenue CAGR, founder CEO flag, and has_sl_data.

We train three models: Ridge Regression (L2-regularized linear baseline), Random Forest (ensemble of decision trees), and Gradient Boosting (sequential trees correcting prior errors). For each horizon we use an 80/20 train/test split and report R2 on the held-out test set. For the 365-day Random Forest, we additionally run 5-fold cross-validation and rolling-window validation (train on three consecutive years, test on the next). The hype versus fundamentals comparison restricts all three models to the same observation set.

3. Results

3.1 Descriptive Statistics and Return Distributions

Figure 2 shows the distribution of winsorized returns at each horizon. Day-0 returns are tightly clustered around 0%, reflecting that many IPOs close near the offer price. The distribution widens substantially at longer horizons, reflecting compound uncertainty.

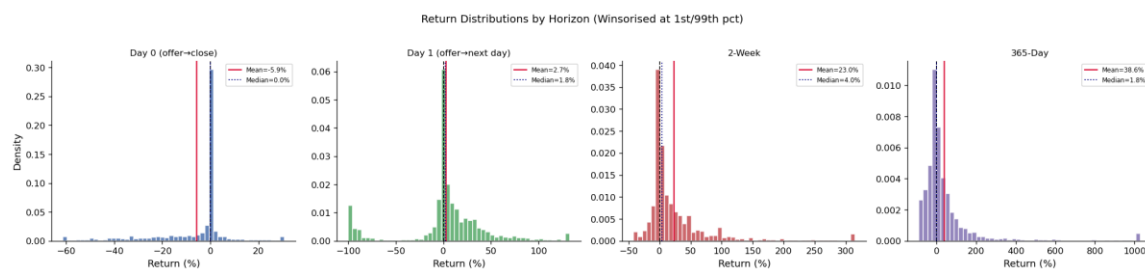


Figure 2: Return distributions by horizon (winsorized at 1st/99th percentile). Red vertical line = mean; blue dotted line = median. Note the widening distribution as the horizon extends.

Figure 3 illustrates the difference between raw returns and BHAR at the 365-day horizon. The raw mean return of 38.6% appears positive, but the median BHAR falls to -11.96%,

meaning the typical IPO underperformed the S&P 500 by about 12 percentage points over its first year, consistent with Loughran and Ritter (1995).

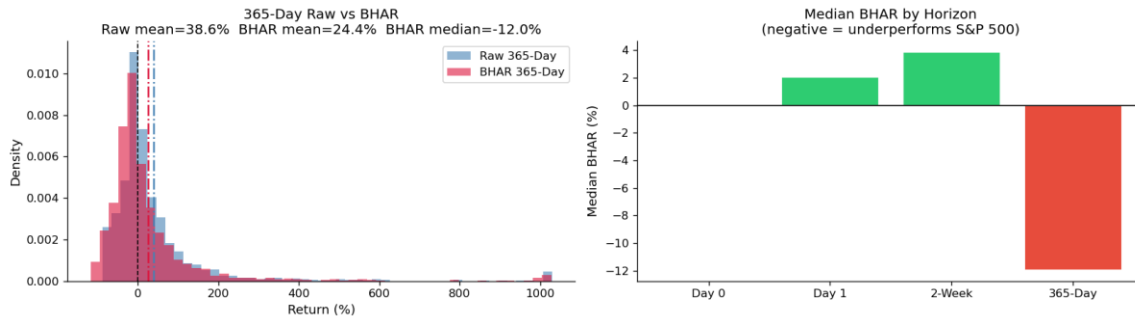


Figure 3: Left panel compares the distributions of raw 365-day returns and 365-day BHAR. Right panel shows median BHAR at each horizon. A negative bar means the typical IPO underperformed the S&P 500 over that period.

3.2 Model Performance

Table 2 reports out-of-sample R^2 for all three models across all four horizons. Random Forest consistently outperforms Ridge and Gradient Boosting. Gradient Boosting produces a negative R^2 at Day 1 (-0.051), likely due to overfitting on the smaller sample (1,716 observations). Figure 4 plots the results visually.

Table 2: Out-of-sample R^2 by model and horizon (80/20 test split)

Model	Day 0	Day 1	2-Week	365-Day	Day 0 (BHAR)	365-Day (BHAR)
Ridge	0.101	0.057	0.159	0.145	0.102	0.145
Random Forest	0.170	0.107	0.222	0.382	0.174	0.384
Gradient Boosting	0.117	-0.051	0.111	0.215	0.122	0.214

Note: R^2 measured on the held-out 20% test set. BHAR columns shown for Day 0 and 365-Day only; patterns are similar for Day 1 and 2-Week.

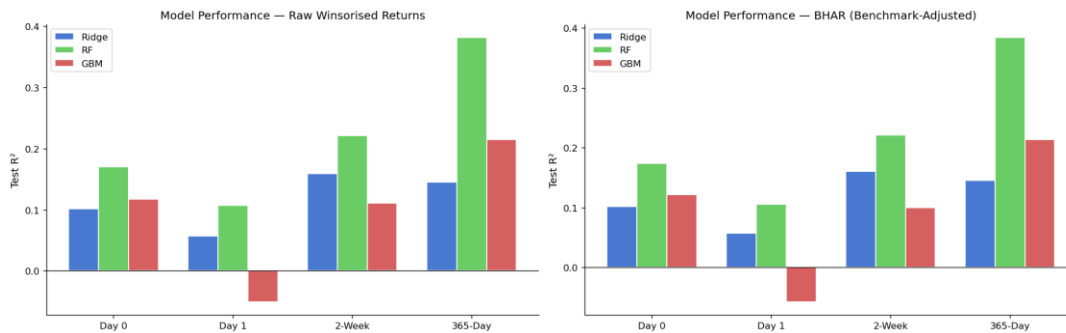


Figure 4: Test R^2 by model and return horizon. Left: raw winsorized returns. Right: BHAR. RF = Random Forest, GBM = Gradient Boosting.

Performance is weakest at Day 1 and strongest at 365 days, where the Random Forest achieves an R^2 of 0.382 on raw returns and 0.384 on BHAR. Five-fold cross-validation gives a mean R^2 of 0.255 (range: 0.094 to 0.423), reflecting sensitivity to which years land in training versus test.

3.3 Feature Importance

Figure 5 shows Mean Decrease Impurity (MDI) for each feature at each horizon. MDI measures how much each feature's splits reduce prediction error across the trees in the Random Forest ensemble.

Log Proceeds dominates at every horizon, with MDI rising from 0.336 at Day 0 to 0.790 at 365 days. Company Age is the second most stable feature. VC-Backed matters at short horizons (0.110 at Day 0) but falls to 0.011 at 365 days, consistent with VC certification generating early enthusiasm that fades over time. The S-1 features show limited predictive power, likely due to the 28.1% coverage rate.

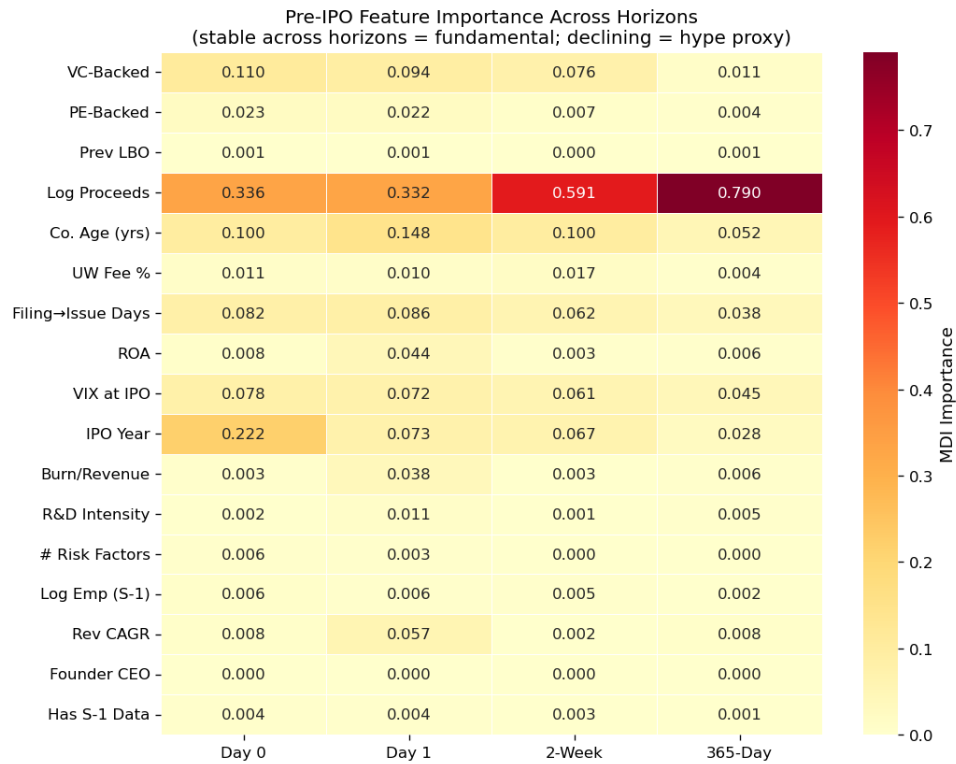


Figure 5: Random Forest MDI feature importance across all four horizons. Darker red = higher importance. Features stable across horizons (Log Proceeds, Company Age) are likely fundamental predictors; features declining at longer horizons (VC-Backed, IPO Year) reflect shorter-run market dynamics.

To complement the MDI analysis, we compute permutation importance on the held-out test set at each horizon. Permutation importance measures the decrease in out-of-sample R^2 when a feature's values are randomly shuffled (Breiman, 2001). Figure 6 displays the results, with the top three features highlighted in red. Log Proceeds, VC-Backed, and IPO Year consistently emerge as the most important, but their relative ordering shifts across horizons. At Day 0, VC-Backed leads, suggesting investor-facing signals dominate early price formation. By the 365-day horizon, Log Proceeds takes a decisive lead, consistent with the hype-to-fundamentals narrative: deal size becomes increasingly relevant as short-run sentiment fades. Features such as Company Age, Revenue CAGR, and ROA gain modest importance at longer horizons, reinforcing the view that fundamentals matter more for long-run returns.

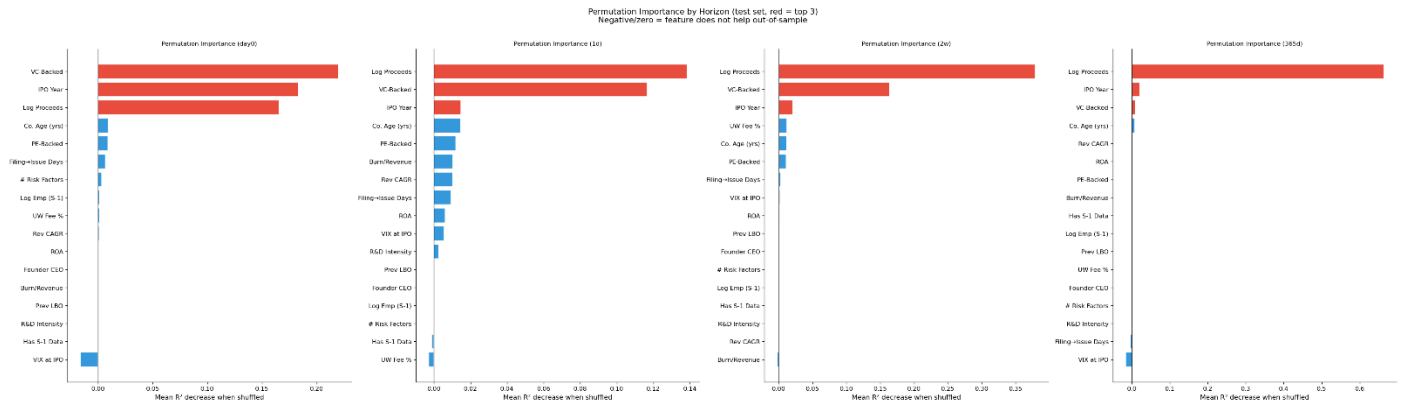


Figure 6: Permutation importance (test set) by horizon. Red bars indicate the top three features at each horizon.

Negative or zero importance means the feature does not help out-of-sample prediction.

3.4 Hype versus Fundamentals

Table 3 and Figure 7 show the results of the formal hype versus fundamentals test at the 365-day horizon. All three models are estimated on the same observation set to ensure comparability.

Table 3: Hype versus fundamentals test (Random Forest, 365-day horizon, common observation set)

Feature Set	365-Day Raw R2	365-Day BHAR R2	Description
Hype only	0.047	0.043	Day-0 and Day-1 returns only
Fundamentals only	0.014	-0.000	Pre-IPO structural features only
Full (both)	0.078	0.096	All features combined

Note: All three models evaluated on the same observations. A negative R2 means the model performs worse than predicting the mean return.

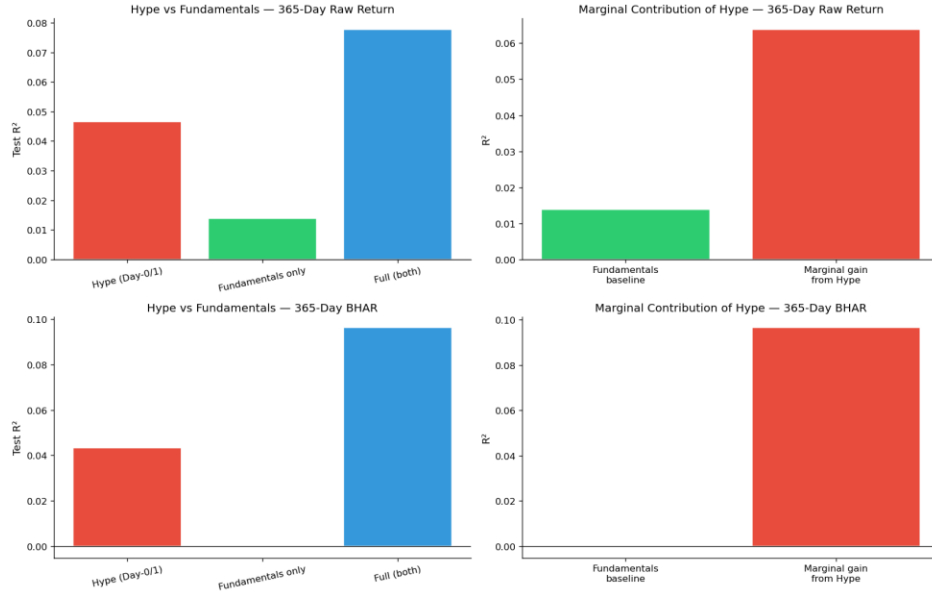


Figure 7: Left panels compare R² across the three model specifications at the 365-day horizon for raw returns (top) and BHAR (bottom). Right panels show the marginal R² gain from adding hype features to the fundamentals baseline.

The results do not support our original hypothesis. The fundamentals-only model achieves an R² of 0.014 for raw 365-day returns and -0.000 for BHAR, adding essentially no predictive power over the mean. The hype model reaches 0.047 (raw) and 0.043 (BHAR). The full model improves on both (raw R² = 0.078, BHAR R² = 0.096), confirming that hype contributes substantially more, likely because Day-0 and Day-1 prices incorporate roadshow and order-book signals not captured in pre-IPO documents.

3.5 Rolling-Window Validation

Figure 8 shows rolling-window validation results for the 365-day BHAR model, where the model trains on three consecutive years and tests on the next.

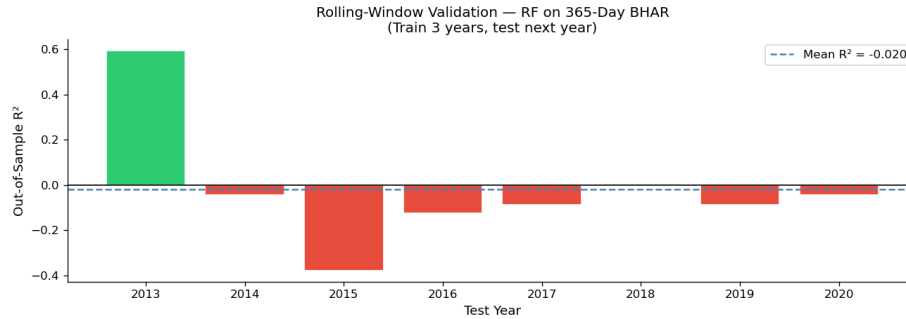


Figure 8: Rolling-window R^2 for the 365-day BHAR Random Forest (train 3 years, test next year). Dashed blue line = mean R^2 .

The 2013 window is the exception, with R^2 of 0.596. In most other years R^2 is near zero or negative, giving a mean of -0.020. This indicates that the single-split R^2 of 0.384 is partly a product of within-period correlation among IPO characteristics, and that year-to-year generalization is not reliable with our current features.

4. Conclusion

This paper predicted U.S. IPO returns at four horizons for 3,158 IPOs from 2009 to 2023. The median 365-day BHAR of -11.96% confirms long-run underperformance (Loughran and Ritter, 1995). Random Forest consistently outperforms, reaching $R^2 = 0.384$ for 365-day BHAR. Log proceeds dominates at all horizons; VC-backing fades over time. The central finding is that pre-IPO fundamentals add essentially no predictive power for 365-day BHAR ($R^2 = -0.000$), while early return signals reach 0.043 and the combined model 0.096, suggesting Day-0 and Day-1 prices incorporate roadshow and order-book information not in pre-IPO documents. Rolling-window validation (mean $R^2 = -0.020$) shows cross-period forecasting remains difficult. Limitations include sparse S-1 coverage (28.1%), U.S.-only scope, and potential log proceeds bias. Future work should improve S-1 matching, extend the framework internationally, and incorporate post-lockup trading signals.

References

- Breiman, L. (2001). Random forests. *Machine Learning*, 45(1), 5-32.
- Loughran, T., and Ritter, J. R. (1995). The new issues puzzle. *Journal of Finance*, 50(1), 23-51.
- Loughran, T., and Ritter, J. R. (2004). Why has IPO underpricing changed over time? *Financial Management*, 33(3), 5-37.
- Meggison, W. L., and Weiss, K. A. (1991). Venture capitalist certification in initial public offerings. *Journal of Finance*, 46(3), 879-903.
- Ritter, J. R. (1991). The long-run performance of initial public offerings. *Journal of Finance*, 46(1), 3-27.
- Sterne, J. A. C., White, I. R., Carlin, J. B., Spratt, M., Royston, P., Kenward, M. G., Wood, A. M., and Carpenter, J. R. (2009). Multiple imputation for missing data in epidemiological and clinical research: Potential and pitfalls. *BMJ*, 338, b2393.
- van Buuren, S. (2018). *Flexible Imputation of Missing Data* (2nd ed.). Boca Raton: Chapman and Hall/CRC.